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CS 499 Milestone Four: Enhancement Three — Databases

Artifact Description

Project Three, the Android inventory management application I developed as an evaluation submission for the CS 360 course, is built using a mobile-adaptive technology stack, and provides core support for functions including inventory registration, stocktaking, and status updates.

Justification for Inclusion in the ePortfolio

I selected this technical artifact to submit for evaluation in the database category. The core reason for this choice is that it proves my ability to work with databases in real mobile application scenarios. This project uses the local SQLite library to store user login and inventory data, and implements table creation, data storage, and CRUD operations, fully meeting all requirements for the evaluation.

To optimize the database structure and processing capacity, the authors of this paper implemented multiple modifications to Java's DatabaseHelper.java class, added new logic including table constraints and duplicate data verification, optimized core operation methods, and effectively improved the database's reliability, security, and code maintainability.

Course Outcomes

This enhancement supports Course Outcome 4 because I used Android Studio, Java, SQLite, and database programming practices to improve a working application. It also supports Course Outcome 5 because I improved safer data handling by adding validation, duplicate checking, and password hashing instead of storing passwords as plain text. This enhancement also connects to Course Outcome 2 because I am explaining my technical work clearly in this narrative and in my ePortfolio.

My outcome coverage plan has not changed. My first enhancement focused on software design and engineering, my second enhancement focused on algorithms and data structures, and this third enhancement focuses on databases. Together, these improvements help show my growth across the main areas of computer science.

Reflection

While completing this enhancement, I learned that database design is very important for application quality and security. The original database worked for basic storage, but it did not fully protect against duplicate records or invalid data. By improving the table constraints and adding validation inside the database helper, I made the app more reliable.

One challenge was making sure the database changes still matched the rest of the application. Since the inventory screen depends on the database methods, I had to keep method names and return types consistent. This helped me understand how database changes can affect the whole application. Overall, this milestone helped me improve my skills in SQLite, CRUD operations, validation, and secure data handling.

Conclusion

Overall, this milestone helped me strengthen the database part of my CS 360 Inventory App. The enhanced database now includes better constraints, duplicate checking, safer CRUD operations, password hashing, and improved organization. These improvements will help me present the artifact professionally in my final ePortfolio and demonstrate my database skills.

Reference

National Institute of Standards and Technology. (2024). *The NIST cybersecurity framework (CSF) 2.0*. U.S. Department of Commerce. <https://doi.org/10.6028/NIST.CSWP.29>